

AI Fundamentals Glossary

This glossary lists definitions for the key terms that you learned in each module. The terms are listed alphabetically.

Term	Definition
Algorithm	An algorithm is a set of rules or a step-by-step procedure that guides machines to perform a task or solve a problem.
AI agent	An AI agent is a tool that uses AI techniques and has agency to act in its environment based on set goals. It can select and perform actions autonomously to achieve those goals, interacting with tools, data, and other systems.
AI ethics	AI ethics refers to a multidisciplinary field that studies how to optimize AI's beneficial impact while reducing risks and adverse outcomes.
Artificial intelligence	Artificial intelligence, or AI, is the ability of machines to perform tasks that usually require human intelligence, such as recognizing patterns, making decisions, or understanding language.
Artificial language	An artificial language is a language that has been deliberately and purposefully created to provide clear, precise instructions to computers.
Capabilities	Capabilities refer to the core functions or skills a system can perform. In the context of AI, these are specific tasks that an AI system is designed to do.
Classical machine learning	Classical machine learning is a group of models where computers use mathematical calculations to identify patterns in data and make suggestions or decisions based on those patterns.
Classification	Classification refers to categorizing or grouping text based on common patterns, such as using certain words or sentence structures.
Computer vision	Computer vision is a field of artificial intelligence that trains computers to process, interpret, and make decisions based on visual information, such as images or videos
Computer vision AI	Computer vision AI is a technology that enables machines to interpret and recognize visual information from digital images and videos using algorithms and pattern recognition.
Confidence score	A confidence score is a numerical value that indicates the model's level of certainty about its perception. For example, a model might indicate it's 85% confident that a photo shows a dog.

Term	Definition
Conversational AI	Conversational AI refers to technologies, such as chatbots or virtual agents, that users can talk to. This interaction with machines feels remarkably similar to how people speak with one another.
Data	Data is information that people collect, store, and use for analysis, decision making, or communication in digital systems.
Data analysis	Data analysis is the process of examining information to identify patterns, draw conclusions, and inform decision making. Data analysis typically involves gathering and organizing data, checking it for errors, and then using tools to interpret the numbers.
Data source	A data source is the original location, system, or means from which you collect information for training a deep learning model.
Decision tree	A decision tree is a classical machine learning model that splits data into branches to reach decisions or classifications.
Decision-making AI	Decision-making AI evaluates multiple possible actions and chooses the optimal action based on factors including efficiency, accuracy, and suggested outcomes.
Deep learning	Deep learning is a type of machine learning that uses artificial neural networks with multiple layers. Deep learning identifies complex patterns by learning simple features first and then combining them into more advanced concepts.
Fairness	Fairness refers to a deep learning model's ability to produce equitable outcomes for individuals and groups by helping minimize bias. Fairness in AI agent decisions is the principle that automated systems should not favor or discriminate against individuals or groups based on characteristics, such as income level, personal attributes, or geographic location.
Generative AI	Generative AI refers to systems that use deep learning models and their datasets to create original content, such as text, images, video, audio, or software code, in response to a user's prompt or request.
Goal-based agent	A goal-based agent is a system that chooses actions by considering which will best help it reach a set goal or wanted outcome.
Hallucination	A hallucination is a response from a foundation model that includes off-topic, repetitive, incorrect, or fabricated content. Hallucinations involving fabricating details can happen when a model is prompted to generate text, but the model doesn't have enough related text to draw upon to generate a result that contains the correct details.

Term	Definition
Hidden data	Hidden data refers to information that exists in an organization's systems or processes but is not actively used or analyzed, typically because it's too difficult to process, organize, or analyze using traditional tools.
Human-in-the-loop	Human-in-the-loop (HITL) refers to a system or process in which a human actively participates in the operation, supervision, or decision making of an automated system.
Image classification	Image classification is a computer process that analyzes a picture and sorts it into categories based on what objects or scenes are present.
Intelligence	Intelligence is the ability to learn, understand, and solve problems using knowledge and reasoning.
Intent recognition	Intent recognition is the process by which a language system determines what a person intends to do or has done based on their words or actions.
Large language model	A large language model (LLM) is a type of AI system trained on massive text datasets, enabling it to understand questions, generate answers, and hold human-like conversations
Learning	Learning is the process of gaining new knowledge, skills, or understanding through experience, study, or instruction.
Learning agent	A learning agent is a type of AI system that improves its actions over time by using feedback from its environment.
Linear regression	Linear regression is a machine learning model that recommends continuous outcomes by fitting a straight line to observed data points.
Logistic regression	Logistic regression is a classical machine learning model that helps solve problems where the goal is to estimate one of two possible outcomes based on input variables.
Machine learning	Machine learning is a subset of artificial intelligence that helps computers learn from experience. With machine learning, computers can perform tasks independently and improve as they accumulate more data.
Model-based reflex agent	A model-based reflex agent is a type of AI tool that uses an internal memory or model to keep track of what has happened in its environment, not just what is happening right now.
Multimodal AI	Multimodal AI creates rich new content from a variety of sources and data types, including text, images, audio, and video.

Term	Definition
Natural language	A natural language is any language that people use to communicate in everyday life.
Natural language processing (NLP)	Natural language processing (NLP) is a subfield of computer science and artificial intelligence that uses machine learning to enable computers to understand and communicate with human language.
Neural network	A neural network is a computer system composed of layers of interconnected units, known as nodes, designed to identify patterns in data and make recommendations or decisions based on that data.
Permissions	Permissions are the specific rights or access levels you give to an AI tool, deciding what actions it can take, what data it can use, and what choices it can make independently.
Predictive AI	Predictive AI involves using statistical analysis and machine learning to identify patterns, anticipate behaviors, and forecast upcoming events.
Prompt	A prompt is the input statement, question, or instruction you provide to a generative AI system to guide it in creating a specific output.
Reasoning	Reasoning is the mental process of applying knowledge and logic to understand information, solve problems, and make informed decisions. Reasoning involves thinking critically, analyzing ideas, and drawing conclusions based on knowledge and experience.
Reinforcement learning	Reinforcement learning is a machine learning model in which a tool learns to make decisions by trying different actions and receiving feedback in the form of rewards or penalties.
Robustness	Robustness refers to a deep learning model's ability to maintain accurate performance when presented with new, noisy, or challenging data.
Safety risks	Safety risks with AI agents occur when these systems cause harm to people or property because of unintended actions, misuse, or technical failures.
Self-supervised learning	Self-supervised learning is a method where an AI system uses parts of its own data to create tasks and learn from them without needing labeled examples.
Semi-supervised learning	Semi-supervised learning is a machine learning model that uses a small amount of labeled data, along with a larger amount of unlabeled data, to train an AI tool.
Sentiment analysis	Sentiment analysis enables AI to detect emotion or tone in language and helps companies analyze customer reviews. For example, a written complaint

Term	Definition
	about slow shipping becomes a data point triggering a customer service response.
Simple agent	A simple agent is a computer program that reacts to its environment using fixed rules, always giving the same response to the same situation. It does not remember past actions or learn from experience.
Supervised learning	In supervised learning, the model trains using labeled examples, allowing it to provide insights for new data.
Supervisory signal	A supervisory signal is the correct output or label provided during training that guides the model in learning the relationship between inputs and outcomes. It serves as feedback in supervised learning, enabling the model to recognize patterns and offer insights and recommendations.
Thinking	Thinking is the process of using information to form ideas, solve problems, or make decisions.
Training	Training is the process of teaching a language model to recognize patterns in language by exposing it to large amounts of text and adjusting its internal settings based on the examples it sees.
Transparency	Transparency refers to a deep learning model's degree of openness about its design, training data sources, and decision-making process.
Unstructured data	Unstructured data is information that doesn't follow a set format or organized structure, making it hard to store in traditional databases. Unstructured data includes text, images, audio, and video that do not fit into rows and columns.
Unsupervised learning	Unsupervised learning is a machine learning model in which the AI tool analyzes data without labeled answers or guidance, looking for patterns or groupings on its own.
Utility-based agent	A utility-based agent is a system that uses a numerical score, called utility, to compare different choices and pick the one with the highest value.